ACTIVITY 4.2 Mendel and heredity

SKILLS
- Literacy
- Processing and analysing data and information

1 Determine whether each statement below is true or false.

	TRUE	FALSE
a A gene is part of a chromosome.	☐	☐
b The human genome is different from the monkey genome.	☐	☐
c A flower has male parts and female parts.	☐	☐
d Mendel crossed different types of dogs to create different breeds.	☐	☐

2 Refer to the textbook for information on Gregor Mendel and his experiments on pea plants. Colour the pea plants in the crossing diagram in Figure 4.1, using green and yellow as appropriate. Write in the resulting colour for each offspring.

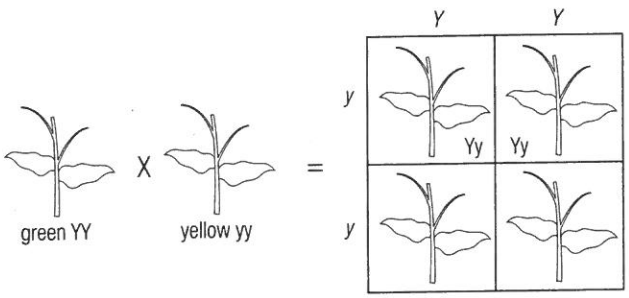


Fig 4.1

3 Describe what is shown in Figure 4.2. Then colour the diagram suitably and complete the Punnett square showing the possible genotypes of the offspring.

_____

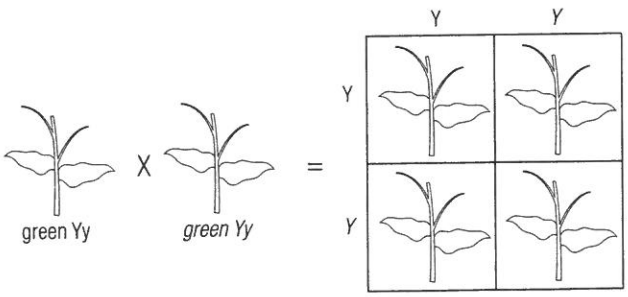


Fig 4.2

Copyright © Oxford University Press ISBN 9780195573626

4 What is Mendel's factor of inheritance?

5 State whether the dominant trait or the recessive trait is expressed when the two are paired.

6 Define the terms 'genotype' and 'phenotype'.

7 Use the words 'dominant' and 'recessive' to explain what occurred when Mendel crossed pure-bred yellow plants with pure-bred green plants.

8 Explain the possible colours of the offspring for Mendel's second generation cross, where the hybrid is crossed with the same hybrid.

Copyright © Oxford University Press ISBN 9780195573626